Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-2630
Project Name	AI Fit Track API
Maximum Marks	3 Marks

Scalability & Future Plan

The team's original ER diagram (see Project Documentation, EPIC 1) was deliberately designed richer than what this cycle implements, so it doubles as the product roadmap:

Planned Entity / Capability	Purpose	Status
WORKOUT_CATEGORY (dedicated collection)	Move category from a fixed schema enum to an admin-manageable collection, allowing new categories without a code change	Future roadmap
EXERCISE (sets/reps breakdown)	Track individual exercises within a workout session (sets, reps, weight) instead of a single aggregate entry	Future roadmap
AI_INTERACTION (log collection)	Persist every Gemini prompt/response pair for history, auditing and to let recommendations improve using the user's past AI interactions	Future roadmap
PROGRESS (tracking collection)	Track body-composition metrics (weight, body fat %) over time and correlate them with workout history	Future roadmap
Frontend (client/ folder)	A React or mobile UI on top of the existing REST API	Scaffolded, not built this cycle
Async AI processing	Move the two Gemini-backed endpoints to a queued/async pattern to smooth out their 5–7.3 s latency for end users	Future roadmap